MILITARY SPECIFICATION

SKINS, FOR M138E3 WARHEAD SECTION

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of skin entitled Skin for the M138E3 warhead section which is used on the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-406 - Plastics, Organic, General Specification, Test Methods.

MILITARY

MIL-Y-1140 - Yarn, Cord, Sleeving, Cloth and Tape Glass.
MIL-R-7575 - Resin Polyester, Low Pressure Laminating.
MIL-F-9118 - Finish for Glass Fabrics.
MIL-R-9300 - Resin, Epoxy, Low Pressure Laminating.
MIL-M-15617 - Mats, Fibrous, Glass For Reinforced Plastics.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

FSC: 1135

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DRAWINGS

ORDNANCE CORPS

8862797 - Skin.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.1.1 Reinforcement. The woven cloth shall be continuous filament type glass fabric and shall be in accordance with specification MIL-Y-1140.

3.1.2 Glass mat. The glass mat shall conform to the requirements of MIL-M-15617.

3.1.3 Finish. The finish shall be compatible with the resin system and shall be accordance with MIL-F-9118.

3.1.4 Resins. Polyester resins shall meet the requirements of MIL-R-7575. Epoxy resins shall meet the requirements of MIL-R-9300.

3.2 Assembly. The skin shall comply with all requirements specified on Drawing (dwg) 886797, and with all requirements specified in applicable specifications.

3.2.1 Laminates construction. The skin shall have the following construction:

GEL Coat - .030 maximum in areas specified on drawing.
1 layer of 1/2 ounce chopped strand mat.
1 layer of 3 ounce chopped strand mat.
1 layer of 10 ounce fabric (EM63 or equal).
1 layer of 3 ounce chopped strand mat.
1 layer of 10 ounce fabric (EM62 or equal).
1 layer of 3 ounce chopped strand mat.
1 layer of 10 ounce fabric (EM63 or equal).

The laminating resin shall meet the requirements of MIL-R-7575.

3.2.2 Bonded joint area. The epoxy resin used in bonding the skin shall meet the requirements of MIL-R-9300.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Bearing strength. The skin shall have a bearing strength of 25,000 PSI minimum when tested as specified in 4.4.1.

3.6 Ultimate tensile strength. The skin shall have an ultimate tensile strength of 1,500 PSI minimum when tested as specified in 4.4.2.

3.7 Process. A detailed description of the manufacturing and fabrication process and methods of control of manufacturing variables shall be submitted by the contractor in the form of a titled, numbered, and dated process specification to the procuring agency for approval. After approval, the process specification shall form a part of this specification and copies shall be made available by the contractor for use by authorized Government inspectors at the contractor's plant. The material and processes in production shall conform to the requirements of the process specification, in addition to the requirements of this specification and shall not be changed in production without approval of a new superseding process specification. The process specification shall also contain identification of all materials employed and specific requirements for resin content, specific gravity, barcol hardness, allowable and non-allowable defects, repairable and non-repairable defects and the method of repairing any repairable defects.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Critical area. The area between station 10 and station 15 and the area between station 133 and 139 of dwg. 8862797 shall be uniform, smooth and free from uncured or unbonded areas, gaps, cracks, holes, blisters, resin pockets, areas lacking resin, tackiness, excess surface resin, laps, wrinkles, de-lamination, air or gas pockets, patches, porosity and other similar defects.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8862797) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Bearing strength (see 3.5)		X
3. Ultimate tensile strength (see 3.6)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Skin (dwg. 8862797).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Gaps between pieces of glass fabric in any lamination	Visual
	102. Lap widths less than one-half inch	Visual
	103. Lap widths more than one inch	Visual
	104. Evidence of foreign material in the finished plastic material	Visual
	105. GEL coat on unspecified surfaces	Visual
	106. Surface finish on mating and bearing surfaces in excess of 125 rms	Visual
	107. Evidence of release agent	Visual
	108. Incorrect color	Visual
	109. Cracks in skin	Visual
	110. Evidence of poor workmanship (see 3.8)	Visual
	111. Objective evidence relating to material missing or improper	Visual
Minor:		
	201. Serialization improper or missing (see 3.4)	Visual

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4.3.2.1.2 Preparation for delivery (see section 5).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Preservation packing and packaging not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Bearing surface	Major
b. Ultimate tensile strength	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Bearing surface and ultimate strength. Specimens from each skin shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.5 and 3.6, the item shall be refected and removed from the lot.

4.3.4 Acceptance inspection equipment. Inspection and testing shall be performed with the acceptance inspection equipment operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Bearing surface. The skin shall be constructed with a minimum of four inches in excess length. Test specimen shall be machined from this excess length and tested in accordance with method 1051 of L-P-406.

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4.4.2 Ultimate tensile strength. Specimen, one inch wide shall be machined from the bonded joint area. The specimen shall be seven inches long with the center of the bonded joint 3.5 inches from the end of the test specimens. the test shall be conducted in accordance with method 1011 of L-P-406, except that the test specimen shall be as specified above and the thickness shall be that of the laminate.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The skin shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The skin shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The skin shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Production or Repair Parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4). Data specified in 4.3.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2376 dated 15 November 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A173

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